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The Scientific method

The Scientific method is a self-correcting process for evaluating ideas with observation & analysis

How do theories become hypotheses?

A Theory explains behaviours or events with a proposed idea that organizes observations

A hypothesis is a testable prediction produced by the theory. A strong hypothesis is falsifiable.

A useful theory {
1. Organizes Observations
2. Can lead to further research
3. implies predictions

that anyone can use to check the theory

that derive practical applications

How do Psychologists gather data without experiments?

Psychologists frequently use experimental methods in research.

However, they also utilize a wide array of non-experimental methods: techniques that observe behaviours without manipulating variables

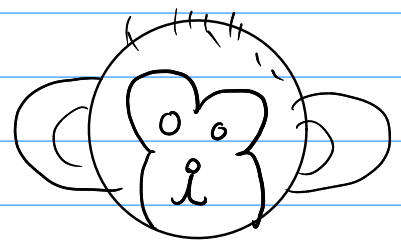
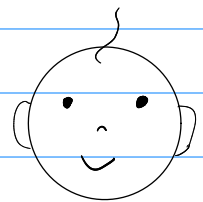
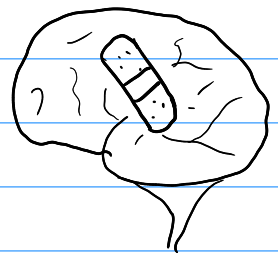
The Case Study

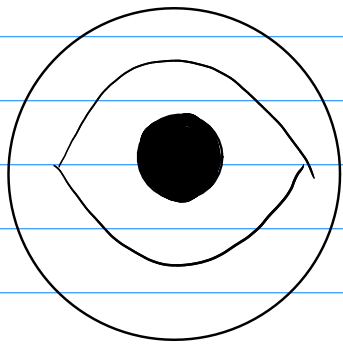
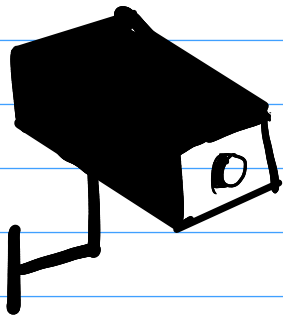
Studies an individual or a group in depth in hope of a generalizable conclusion.

Examples:

- ① Our early knowledge of brain damage come from individuals who were impaired from damage to their brain
- ② Jean Piaget made significant contributions to developmental psychology after carefully observing and questioning a small number of children
- ③ Studies of various animals have taught us a lot about animal intelligence

However, case studies can often suffer from their small sample size





Naturalistic Observations

Recording responses in natural environments

Until recently, this was confined to small-scale projects.

Now however, data from the mass surveillance system gives psychologists the ability to perform massive naturalistic observations

For example, scientists tracked the locations of 15 million people to measure compliance with stay-at-home lockdown regulations

Like Case Studies, naturalistic observations describe not explain.

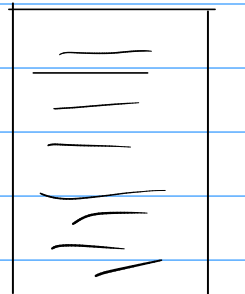
The Survey

A survey looks at a large # of cases in less depth, by asking people to self-report answers to questions.

Researchers have to be careful how they word questions.

Social desirability bias and Self-report bias can affect.

Surveys should try to avoid Sampling Bias by seeking a Random Sample from the population.



What happens after the experiment?

When scientists write a paper, it gets sent to **peer reviews** to get a more objective perspective.

When documenting their findings scientists use precise **operational definitions** so that other scientists can accurately **replicate** the study.

— Personal Response

I tinker with my phone & laptop at the system level so things go wrong a lot. When this happens, I'll make a general **theory** of the cause, and reword that into a more specific **hypothesis** with **operational definitions** that will get me good search engine results.

MCQ:

1. A
2. A
3. B
4. A
5. D
6. A
7. C
8. C